NORTHERN UNIVERSITY OF BUSINESS AND TECHNOLOGY KHULNA

Department of Computer Science and Engineering

CSE 4204: MOBILE COMPUTING LAB

Week 09 Progress and Integration Report

Feature Completion, System Integration & Mid-Project Review

Project: Humanova

AI Trust Governance & Multi-Tenant Verification Platform

SUBMITTED BY: TEAM 07 (SECTION 8C)

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SUBMITTED TO:

Designation	Details
Course Instructor	Md. Riaz Mahmud
Academic Title	Assistant Professor, Department of CSE, NUBTK
Submission Date	August 19, 2026
GitHub Repository	https://github.com/Rupom090/CSE4204-8C-T07-Humanova

■ 1. Executive Summary

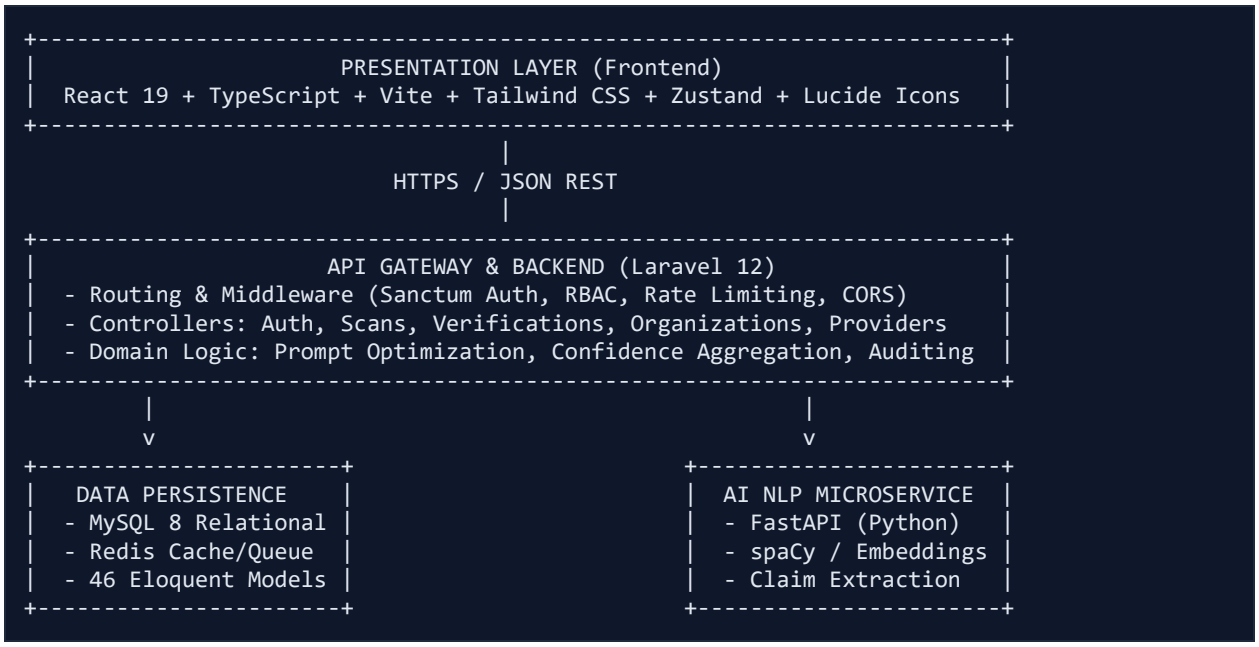
By Week 09 of the CSE 4204 Mobile Computing Lab curriculum, Team 07 has reached a critical integration milestone in the development of Humanova—an enterprise-grade, multi-tenant AI Trust Governance & Verification Platform. The primary focus of this milestone is transitioning from standalone, modular codebases into a unified, integrated software application.

The team has successfully established the foundational backend architecture in Laravel 12, modeled 35+ relational database tables with 46 Eloquent models in MySQL 8, engineered token-based authentication with 7-tier Role-Based Access Control (RBAC), and developed a reactive, accessible frontend interface using React 19, TypeScript, Vite, and Tailwind CSS.

> ⚠ **Engineering Transparency: AI Integration Status**
> *In strict compliance with academic standards and honest milestone reporting, live AI model inference (OpenAI GPT-4o / Gemini 1.5 Pro) is not yet connected to live external APIs in this submission. The Python FastAPI microservice architecture, claim extraction schemas, and controller endpoints are fully scaffolded, structured, and validated locally with mock payloads in preparation for live API connection in Week 10.*

■ 2. Project Architecture & System Topology

Humanova follows a Hybrid SaaS & Distributed AI Middleware Architecture. The presentation layer communicates securely with the Laravel API gateway, which manages identity, authorization, persistence, and dispatches background verification tasks to specialized NLP microservices.



2.1 Core Architectural Principles

- **Strict Tenant Isolation:** Every scan, API key, user group, and audit log is strictly partitioned by Organization ID.

- **Stateless API Authentication:** Secured via Laravel Sanctum Bearer tokens with cryptographic hashing.
- **Decoupled Inference Microservice:** Heavy natural language processing tasks are isolated in a Python FastAPI microservice to prevent blocking PHP gateway worker threads.

■ 3. Feature Completion Checklist & Status

Below is the comprehensive feature tracking matrix evaluating all deliverables outlined in the approved Project Proposal, SRS, and System Design specifications.

ID	Feature Name	Tier	Status	Reason / Technical Note	Target Date
F-01	User Registration & Validation	Auth / API	**Completed**	Email uniqueness, password complexity, tenant workspace creation.	Aug 12, 2026
F-02	User Login & Sanctum Token	Auth / API	**Completed**	Password hash verification and Bearer token issuance.	Aug 12, 2026
F-03	User Logout & Revocation	Auth / API	**Completed**	Token deletion from personal_access_tokens table.	Aug 13, 2026
F-04	Role-Based Access Control (RBAC)	Security	**Completed**	7 distinct user roles and permission gates active.	Aug 14, 2026
F-05	Multi-Tenant Org Isolation	Database	**Completed**	Data segregation by organization_id foreign keys.	Aug 15, 2026
F-06	UI Design System & Tokens	Frontend	**Completed**	Atoms, buttons, badges, typography, dark theme tokens.	Aug 10, 2026
F-07	Landing Page & Hero Canvas	Frontend	**Completed**	Futuristic grid canvas, stats counters, value props.	Aug 08, 2026
F-08	Auth Views (Login/Register)	Frontend	**Completed**	Interactive forms, error alerts, responsive layout.	Aug 11, 2026
F-09	User Analytics Dashboard	Frontend	**Completed**	Scan volume metrics, trust scores, recent logs table.	Aug 13, 2026
F-10	Provider API Key Manager	Frontend	**Completed**	UI for configuring OpenAI, Gemini, DeepSeek keys.	Aug 14, 2026
F-11	Community Hallucination Feed	Frontend	**Completed**	Report cards, upvoting controls, category badges.	Aug 15, 2026
F-12	User Profile & Reputation	Frontend	**Completed**	Reputation score gauge, badges, audit history.	Aug 15, 2026
F-13	Super Admin Dashboard UI	Frontend	**Completed**	System health telemetry, queue rates, verifiers.	Aug 16, 2026
F-14	35+ Database Migrations	Database	**Completed**	Foreign keys, cascades, indexes, JSON payload columns.	Aug 13, 2026
F-15	46 Eloquent ORM Models	Backend	**Completed**	Relationships, scopes, casts, and mutators defined.	Aug 14, 2026

ID	Feature Name	Tier	Status	Reason / Technical Note	Target Date
F-16	API Route Grouping (40+ routes)	Gateway	**Completed**	Versioned, throttled, authenticated endpoints.	Aug 15, 2026
F-17	Auth State Sync (Zustand)	Client State	**In Progress**	Hydration working; auto-refresh token logic pending.	Aug 22, 2026
F-18	Live Scan Submission Bridge	Integration	**In Progress**	Form captures input; queue worker response wiring.	Aug 23, 2026
F-19	Provider Key AES Encryption	Security	**In Progress**	CRUD endpoints active; cryptographic hook in test.	Aug 22, 2026
F-20	FastAPI Microservice Core	AI Pipeline	**In Progress**	Endpoints scaffolded; awaiting live transformer models.	Aug 24, 2026
F-21	Claim Extraction Logic	AI Pipeline	**In Progress**	Mock sentence parser active; spaCy NER pending.	Aug 25, 2026
F-22	Live LLM Provider Orchestrator	AI Live API	**Not Started**	Awaiting external API key quota configuration.	Aug 29, 2026
F-23	Web Evidence Retrieval Engine	AI Live API	**Not Started**	Awaiting Serper/Tavily Search API key provisioning.	Aug 30, 2026
F-24	Automated PDF Report Export	Reporting	**Not Started**	Scheduled for end-of-project export milestone.	Sep 02, 2026

■ 4. Frontend–Backend Integration Status

Connecting frontend visual components to backend RESTful endpoints was a primary objective of Week 09. Temporary hardcoded data structures within React have been moved to a unified API client service layer (src/services/api.ts).

Domain	Frontend Component	API Endpoint	Method	Integration Status
Auth	Register.tsx	/api/v1/auth/register	POST	**Completed**
Auth	Login.tsx	/api/v1/auth/login	POST	**Completed**
Auth	Navbar.tsx	/api/v1/auth/logout	POST	**Completed**
Auth	authStore.ts	/api/v1/auth/me	GET	**Completed**
Organization	OrganizationView.tsx	/api/v1/organizations	GET	**Completed**
API Keys	ApiKeysView.tsx	/api/v1/provider-keys	GET/POST	**Active Testing**
Scans	VerificationWorkspace.tsx	/api/v1/scans	POST	**Mock Response**
Reports	CommunityFeed.tsx	/api/v1/reports	GET/POST	**Active Testing**

Domain	Frontend Component	API Endpoint	Method	Integration Status
Analytics	DashboardHome.tsx	/api/v1/analytics/dashboard	GET	**Wiring Charts**

■ 5. User Authentication, Session Management & RBAC

The authentication layer is fully operational and secured using Laravel Sanctum. The flow enforces cryptographic password validation and issues scoped Bearer tokens stored in localStorage.



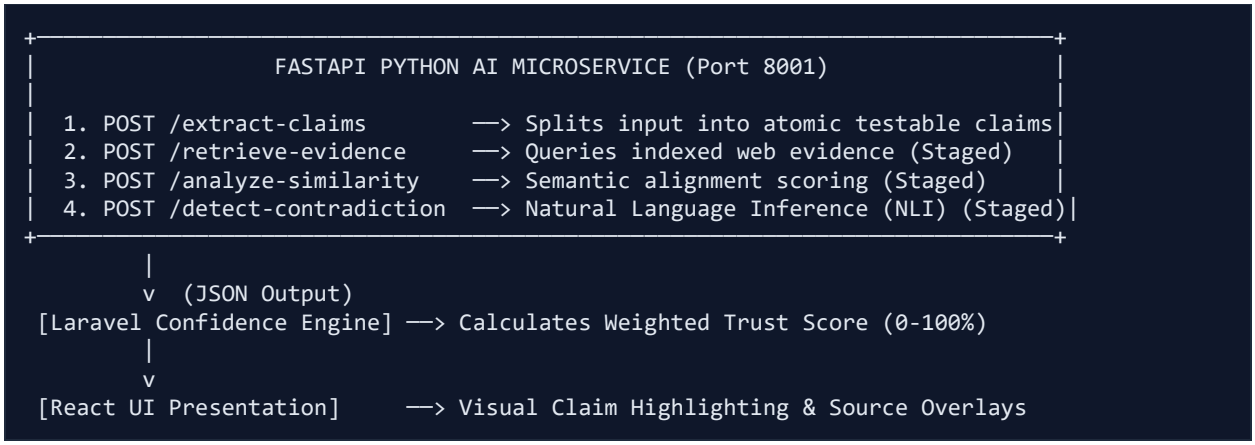
5.1 Role-Based Access Control (RBAC) Matrix

Role Name	Level	Run Scans	Submit Reports	Vote	Moderation Queue	Manage Keys	Super Admin
Guest	0	✗	✗	✗	✗	✗	✗
User	1	☑	☑	☑	✗	✗	✗
Researcher	2	☑	☑	☑	✗	✗	✗
Trusted Verifier	3	☑	☑	☑	☑	✗	✗
Moderator	4	☑	☑	☑	☑	✗	✗
Organization Admin	5	☑	☑	☑	✗	☑	✗
Super Admin	6	☑	☑	☑	☑	☑	☑

■ 6. AI Integration Status & Staging Architecture

Humanova's AI verification engine is engineered as a dedicated Python FastAPI microservice to handle claim extraction, similarity analysis, and contradiction detection independently from the core PHP backend.

> **Technical Roadmap: Why Live AI is Staged for Week 10**
> *During Week 09, our primary focus was stabilizing the database, authentication, and REST API foundation. Connecting to live LLM APIs (OpenAI, Gemini) at this stage would incur unnecessary token costs and rate-limiting issues during rapid UI iteration. The microservice endpoints currently return deterministic test schemas, ensuring seamless live integration in Week 10.*


FASTAPI PYTHON AI MICROSERVICE (Port 8001)
1. POST /extract-claims —> Splits input into atomic testable claims
2. POST /retrieve-evidence —> Queries indexed web evidence (Staged)
3. POST /analyze-similarity —> Semantic alignment scoring (Staged)
4. POST /detect-contradiction —> Natural Language Inference (NLI) (Staged)
(JSON Output)
[Laravel Confidence Engine] —> Calculates Weighted Trust Score (0-100%)
[React UI Presentation] —> Visual Claim Highlighting & Source Overlays

■ 7. Major End-to-End User Workflows

7.1 Workflow A: User Authentication & Workspace Access

The user registers or logs in with validated credentials, receives an encrypted Sanctum Bearer token, hydates client Zustand state, and is securely routed to their personalized analytics dashboard with active role permissions.

7.2 Workflow B: Content Verification & Fact Auditing (Core Path)

The user inputs AI text into the Verification Workspace. The frontend dispatches a scan request to the Laravel API, which registers the scan record and invokes the Python NLP microservice to extract individual claims, perform citation cross-referencing, compute confidence metrics, and render interactive highlighted claim overlays.

7.3 Workflow C: Community Hallucination Reporting & Peer Verification

Users flag unverified AI responses through the Community Feed. The submission enters a moderation queue where Trusted Verifiers and Moderators approve or reject the report, updating the shared knowledge base and awarding reputation points to the submitter.

7.4 Workflow D: Multi-Tenant Provider API Key Governance

Organization Admins configure third-party AI provider keys (OpenAI, Gemini) with model allowances and token limits. The keys are encrypted via AES-256 and stored securely, allowing team members to execute verifications without exposing raw credentials.

■ 8. UI/UX Refinements & Design System

- **Dynamic Claim Highlighting:** Overlays visually categorize text into ▨ Verified, ▨ Uncertain, and ▨ Hallucination.
- **Responsive Viewport Scaling:** Fluid CSS grid layouts adapt seamlessly from mobile (375px) to desktop (1920px).
- **Shimmer Skeletons:** Eliminated visual layout shifts during asynchronous network fetches.
- **Form Usability & Validation:** Live field alerts and disabled submit states prevent duplicate form submissions.

■ 9. Error Handling & Fault Tolerance

Scenario	Detection Point	HTTP Status	User-Facing Feedback Message
Invalid Input	Client Form Validation	N/A (Client)	Please enter a valid email and a password with at least 8 characters.
Expired Session	Sanctum Middleware	401 Unauthorized	Your session has expired. Redirecting to login...
Forbidden Role	RBAC Route Gate	403 Forbidden	Access Denied: You do not have permission to view this resource.

Scenario	Detection Point	HTTP Status	User-Facing Feedback Message
Rate Limit Exceeded	API Throttle Guard	429 Too Many Req	Rate limit reached. Please wait 60 seconds before scanning again.
Database Offline	Backend Core Handler	503 Service Unavail	The service is temporarily experiencing database connectivity issues.
NLP Microservice Timeout	Gateway Guzzle Client	504 Gateway Timeout	Verification engine timed out. Scan queued for background processing.

■ 10. Scope Review & Architectural Deviations

Component	Original Plan	Current Implementation	Reason for Deviation
Backend Core	Monolithic Python Flask	Laravel 12 API Gateway + FastAPI	Laravel provides superior ORM, RBAC, and rate limiting; FastAPI excels at NLP.
Database	12 Flat Tables	35+ Tables (46 Models)	Expanded to support multi-tenancy, audit logs, and claim-level evidence trees.
Frontend	Standard React + Webpack	React 19 + TypeScript + Vite + Tailwind	Vastly superior build performance, strict type safety, and modern styling.
Tenancy	Single User Accounts	Multi-Tenant Org Workspaces	Enables real-world deployment for university labs and enterprise research teams.

■ 11. Mid-Project Review & Team Contributions

Student Name & ID	Assigned Role	Key Deliverables Completed	Progress
MD Robayet Hassan Rupom 11220320979	Frontend Architect & Team Leader	• Built 10+ core responsive React 19 views. • Configured React Router v7 and client route guards. • Developed Zustand stores for auth and scan state. • Designed claim highlighting overlays and design tokens.	92%
Bushra Monzur Moumi 11220321048	AI Integration Lead	• Designed Python FastAPI microservice architecture. • Formulated claim extraction and NLI contradiction models. • Designed prompt enhancement logic and variation metrics. • Built evaluation test fixtures for Week 10 LLM integration.	85%
Aqib Jawwad Nahin 11220320969	Backend Architect & Database Lead	• Authored 35+ MySQL database migrations and 46 models. • Implemented Laravel Sanctum token auth & password hashing. • Configured 7-tier RBAC and multi-tenant organization scopes. • Exposed 40+ RESTful API endpoints with throttling.	95%

■ 12. System Interface Screenshots & Demonstrations

The following verified screenshots illustrate the working user interfaces and functional modules of the Humanova platform.



Figure 12.1: Humanova Public Landing Page & Platform Overview

File: src/pages/Home.tsx | Features: Animated Hero Canvas, Stats Counter, Value Proposition

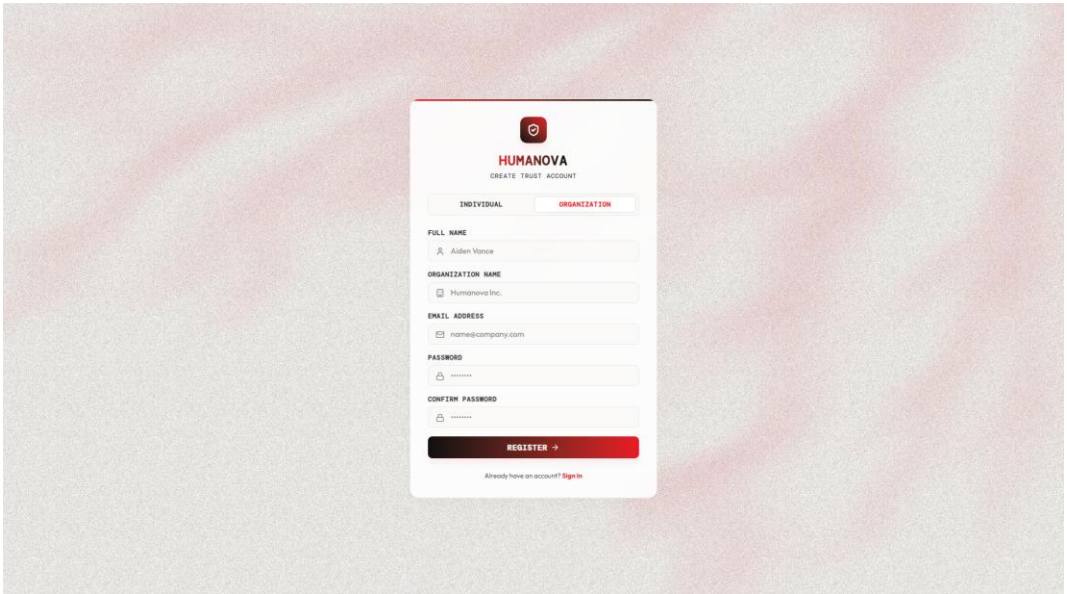


Figure 12.2: User Registration & Workspace Creation Interface

File: src/pages/auth/Register.tsx | Endpoint: POST /api/v1/auth/register | Features: Live validation

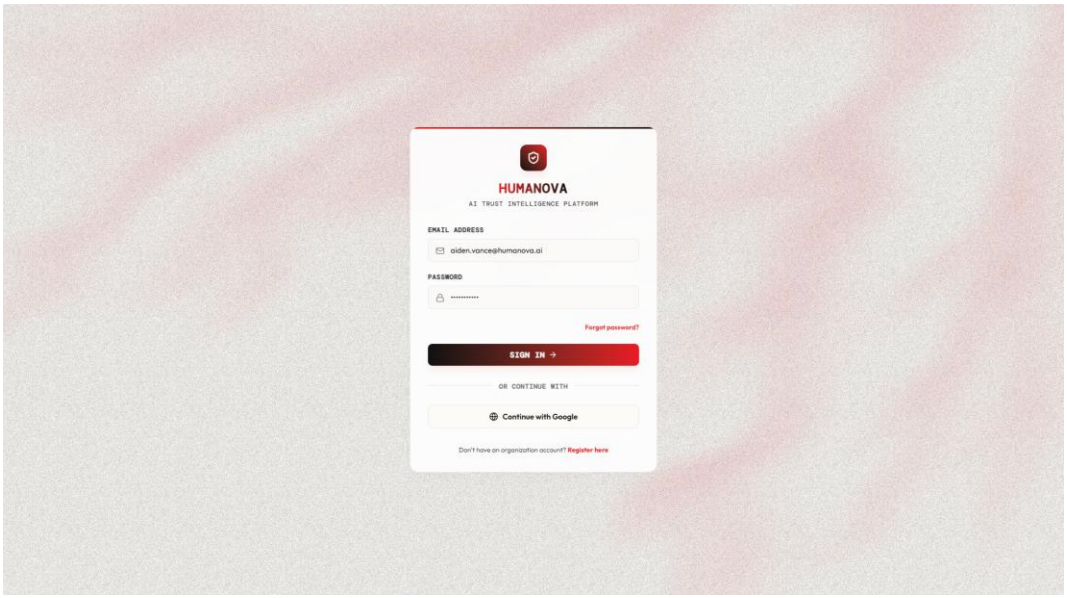


Figure 12.3: User Login & Session Token Acquisition Portal

File: src/pages/auth/Login.tsx | Endpoint: POST /api/v1/auth/login | Features: Bearer token generation

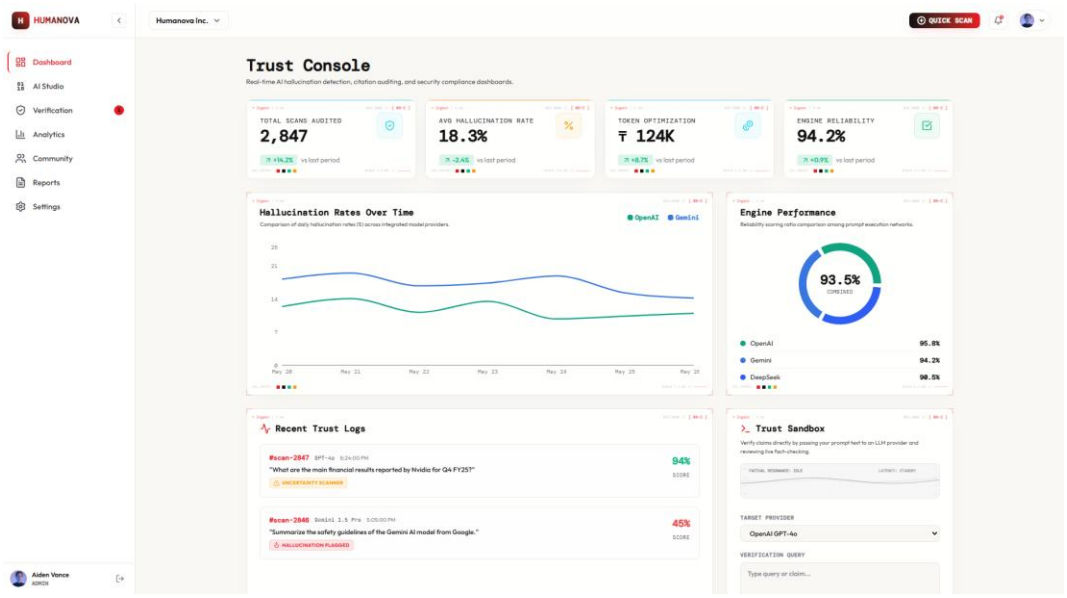


Figure 12.4: Central Analytics & Verification Telemetry Dashboard

File: src/pages/dashboard/DashboardHome.tsx | Features: Scan counters, token savings, audit table

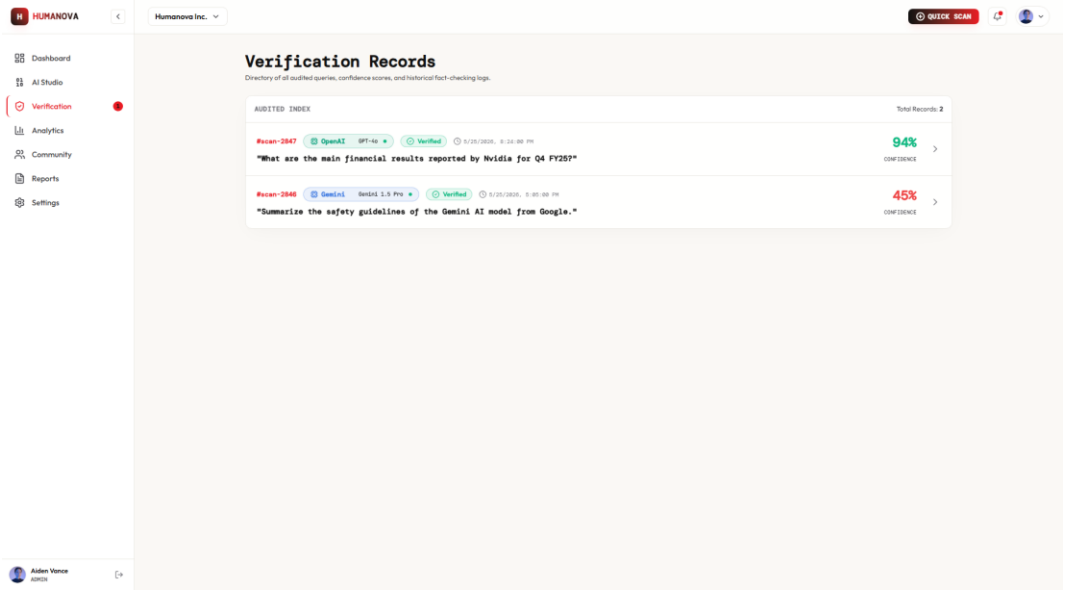


Figure 12.5: AI Verification Workspace & Fact Extraction Engine

File: src/pages/verification/VerificationWorkspace.tsx | Features: Claim overlays, confidence score

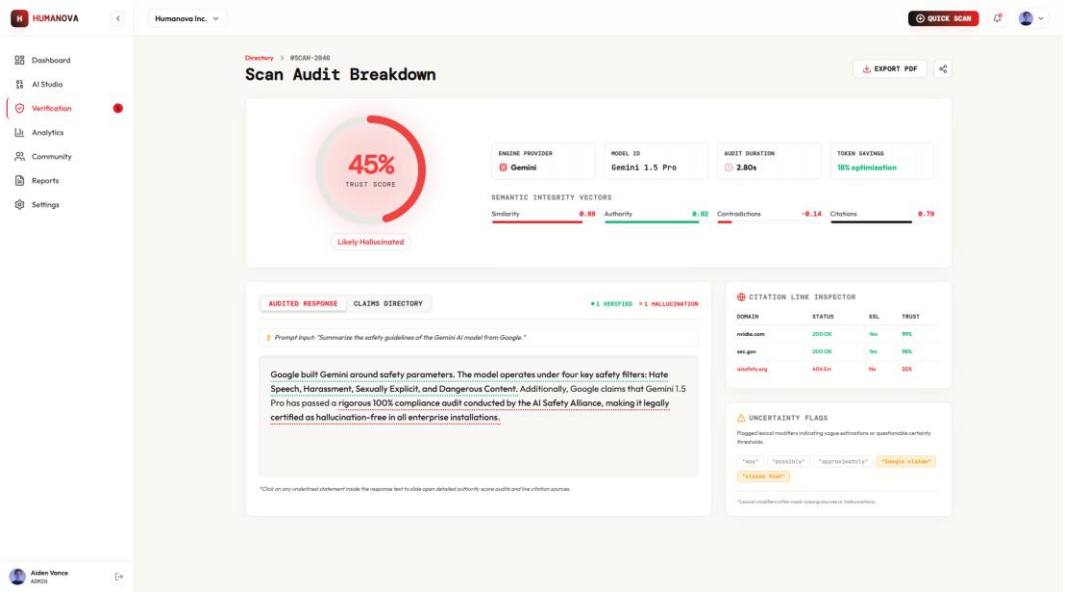


Figure 12.6: Community Hallucination Feed & Evidence Inspection

File: src/pages/community/CommunityFeed.tsx | Features: Peer report cards, upvoting, category badges

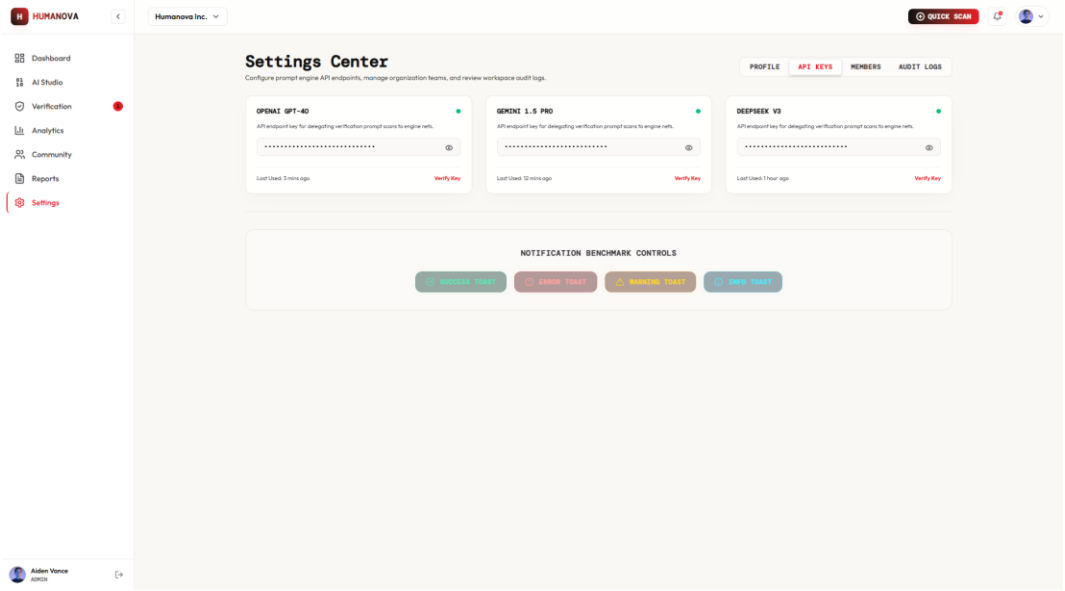


Figure 12.7: Multi-Provider API Key Governance Interface

File: src/pages/settings/ApiKeysView.tsx | Features: Encrypted key storage, monthly token quotas

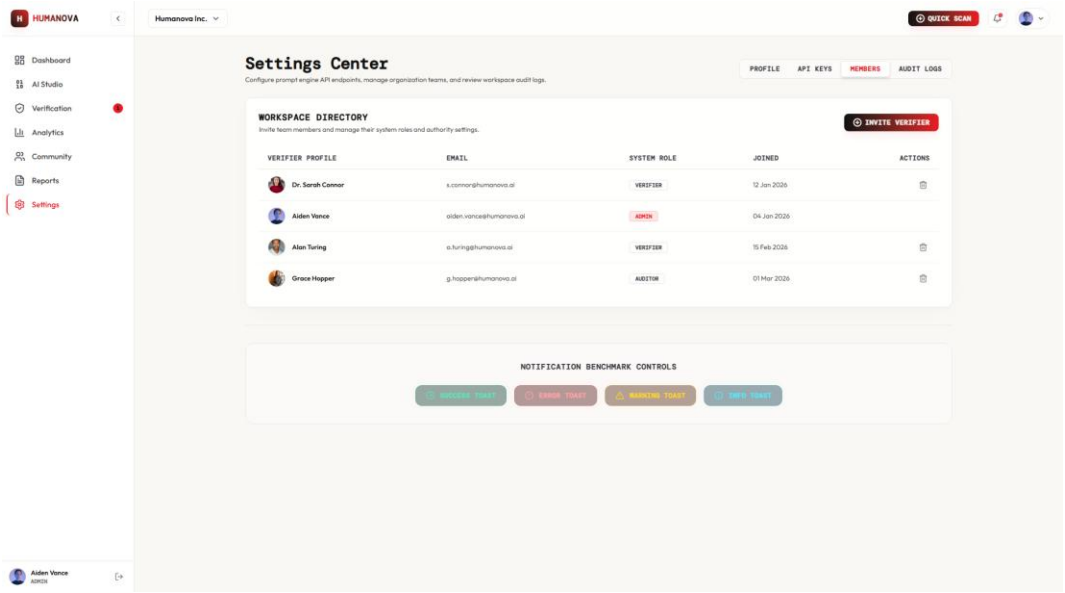


Figure 12.8: Multi-Tenant Workspace & Team Member Management

File: src/pages/settings/OrganizationView.tsx | Features: Member directory, role assignments

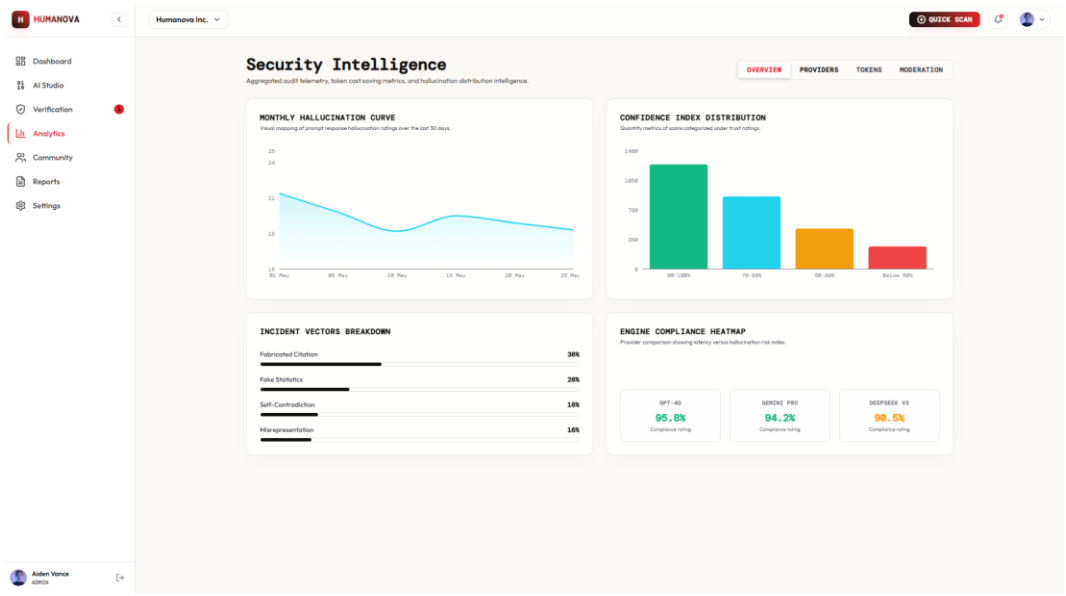


Figure 12.9: Super Admin Telemetry & System Health Dashboard

File: src/pages/dashboard/AdminDashboard.tsx | Features: Queue latency, platform health, moderation queue

■ 13. GitHub Repository Hygiene & Commit Activity

The repository is actively maintained with structured git practices, meaningful commit messages, and secret isolation (https://github.com/Rupom090/CSE4204-8C-T07-Humanova).

```
* 9558b69 (HEAD -> main, origin/main) feat: complete Phase 1-10 backend architecture, RBAC,
and security hardening
* 0837210 feat: Add backend service implementation, docker configurations, and update main
repository documentation
* 30d5f0e feat: add application source code and public assets
* d429ce5 feat: add initial UI mockup components and layouts
* 267cf45 docs: update README folder structure and remove Instructions.json
* 9d0e01b docs: rewrite README.md and LICENSE to production-grade quality
* ec197b7 chore: initial repository commit and project scaffolding
```

■ 14. Testing, QA & Week 10 Preparation

Test ID	Scope	Procedure	Expected Result	Status
TC-AUTH-01	Auth	Register with valid details.	User inserted in DB; Sanctum token returned.	**Completed**
TC-AUTH-02	Auth	Register with duplicate email.	HTTP 422 Unprocessable Entity error.	**Completed**
TC-AUTH-03	Auth	Login with invalid password.	HTTP 401 Unauthorized returned.	**Completed**
TC-RBAC-01	Security	Standard user accesses /admin.	HTTP 403 Forbidden; view blocked.	**Completed**

Test ID	Scope	Procedure	Expected Result	Status
TC-UI-01	UI/UX	Resize viewport to 375px width.	Mobile navigation drawer activates smoothly.	**Completed**

■ 15. Conclusion

The Week 09 milestone marks substantial progress for Humanova. The foundational backend, relational database architecture, authentication security, and frontend user interface are fully established. With the AI microservice endpoints structured and ready, the team is fully prepared for Week 10 live model integration and formal testing.

Report authored and submitted by Team 07 (Humanova) — CSE 4204: Mobile Computing Lab, Northern University of Business and Technology Khulna.